not add value. Even today, however, index funds account for less than 15% of the mutual fund market.[56]

Vanguard was a new *industrial organization* (well, financial organization, but I use the term following the branch of economics specializing in firm structures). Rose-Ackerman (1996) argues that ideologues are drawn to nontraditional corporate structures (especially nonprofits, although Vanguard is not a nonprofit) to share a vision. That vision was more than an index fund; it was a new investment philosophy—aiming for the average, doing it cheaply, and offering it directly to investors. Not for nothing are the followers of Bogle and fans of Vanguard called "Bogleheads."

The Next Generation of Factor Funds

Investors would be better off by gaining access to dynamic factor risk premiums much more cheaply. We need a new generation of factor (index) funds, which are low-cost versions of the factors in HFs. The factors can serve as benchmarks, to ensure that asset owners are getting better value and alleviating agency costs (see chapters 14 and 15). HFs will still exist, and the best HFs will deliver more than factor risk, but we will have upped the standards for HF performance. Given the large left-hand tail losses of these factor risks, greater transparency would greatly benefit investors in optimally allocating to these factor strategies.

The innovation in today's industry and the growing acceptance of factor investing makes this a prime time for new, cheap factor funds. In 2013, we don't see anything resembling a low-cost S&P 500 index fund for the dynamic factors considered in chapter 7. It took decades for cheap equity index funds to go mainstream from their academic beginnings (some findings in academia take a long time to become widely adopted in industry). Professors invented multifactor models in the late 1970s (Ross 1976), but empirical work on factors didn't permeate journals until the 1990s, so the timeline is about right for low-cost multifactor index funds to be introduced today. We can draw some lessons from the introduction of the first factor fund—the equity index fund:

1. The first versions could be failures, even if the idea is right.

   We need to experiment, and anticipate failure.
2. It will likely be introduced and popularized by an outsider.

   It takes a brave HF firm to introduce a new, cheap product that makes old, expensive ones look bad. Or it takes a HF firm that truly has the skill to do it, but those funds are giving up high-margin businesses to move into a low-margin one. Small and startup firms will likely be the pioneers. It may not, however, be the first mover that eventually makes factor portfolios go mainstream.

[56] See Pástor and Stambaugh (2012b).